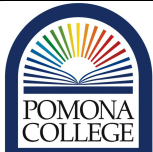


|                                                                                   |                                                         |                             |           |
|-----------------------------------------------------------------------------------|---------------------------------------------------------|-----------------------------|-----------|
|  | Environmental Analysis Teaching and Research Laboratory | Date: 6/16/2025             | SOP No. X |
|                                                                                   | Standard Operating Procedure                            | Title: Nitrate and Ammonium |           |
|                                                                                   | Approved By: Los Huertos                                | Revised: June 30, 2025      |           |

## 1. Scope and Application

## 2. Health and Safety

## 3. Personnel & Training Responsibilities

Students using this SOP should be trained for the following SOPsa:

- SOP1
- SOP2

## 4. Required Materials

4.1 Reciprocating shaker

4.2 Graduated cylinder or repipet type dispenser

4.3 Filter funnels

4.4 Whatman 42 filter paper

4.5 Paper bags or aluminum weigh boats

4.6 Specimen containers, solo cups or nalgene bottles

4.7 Sample cups and caps

4.8 Pipet

4.9 Top-loading balance

## 5. Creating KCL solution

5.1 Add 298.2 g  $\text{CaCl}_2$  to 2 liters distilled water

### 5.2

Lab Procedure

5.3 Fill volumetric container w/ 40 ml solution

5.4 Add 10g of soil into container until solution reaches 50 ml

Table 1: Soil Texture Correction Factor

| Texture | Moist Soil | Dry Soil |
|---------|------------|----------|
| Sand    | 2.3        | 2.6      |
| Loam    | 2.0        | 2.5      |
| Clay    | 1.7        | 2.2      |

**5.5** Cap and shake using shaker for 2 hours

**5.6** Allow sample to sit and soil particles to settle (around 10 mins, no more than 1hr)

**5.7** Place filter paper inside funnels, and set up funnels to drain into labeled containers

**5.8** Pour shaken solution into funnels, don't fill funnels above filter paper line

**5.9** Allow solution to drain into new container until only soil (no liquid) remains in original container

**5.10**

**5.11** compare strip to standard color chart at 30 and 60 seconds

**5.12** Interpretation of N testing quick strips

**5.13** Calculate correction factor using soil textures (average values) See Texture Analysis SOP.

**5.14** Corrected value = test strip value of N (ppm) / correction factor

**5.15** record nitrate ppm and correction factor in field notebook

## 6. References

- 6.1** APHA, AWWA. WEF. (2012) Standard Methods for examination of water and wastewater. 22nd American Public Health Association (Eds.). Washington. 1360 pp. (2014).